

ROCHESTER SKYHAVEN, NH, USA

WMO: 726056

Lat: 43.278N

Lon: 70.922W

Elev: 99

StdP: 100.15

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54791

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.6	(c) -14.9	(d) -25.9	(e) 0.4	(f) -15.9	(g) -23.1	(h) 0.5	(i) -13.1	(j) 10.3	(k) -6.2	(l) 9.2	(m) -6.4	(n) 3.1	(o) 330	(p) 0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	32.0	22.6	30.2	21.7	28.6	20.9	24.1	29.3	23.1	28.0	22.2	26.7	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	17.4	26.7	21.7	16.6	25.7	20.9	15.7	24.8	73.0	29.3	69.1	28.0	65.6	26.5	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.7	6.6	-21.3	34.7	2.6	1.2	-23.2	35.5	-24.8	36.2	-26.2	36.9	-28.2	37.7
			-21.9	25.7	2.5	0.8	-23.7	26.3	-25.1	26.8	-26.5	27.2	-28.3	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-4.3	-3.1	1.2	7.8	13.5	18.3	21.9	20.9	16.9	10.4	4.3	-1.2
	DBStd	10.13	5.95	5.32	5.25	4.71	4.32	3.89	2.99	3.02	3.94	4.30	4.66	5.00
	HDD10.0	1769	443	366	277	97	13	1	0	0	1	48	178	347
	HDD18.3	3718	702	599	531	317	161	48	5	10	74	249	419	604
	CDD10.0	1391	1	1	4	31	122	249	368	339	206	59	8	1
	CDD18.3	298	0	0	0	2	12	47	115	91	30	2	0	0
	CDH23.3	2704	0	0	6	39	173	442	1013	763	252	17	0	0
	CDH26.7	833	0	0	1	11	52	133	333	231	71	1	0	0
Wind	WSAvg	2.4	2.7	2.8	3.1	2.9	2.4	2.0	1.7	1.7	1.8	2.2	2.4	2.5
Precipitation	PrecAvg	1246	87	83	120	110	102	103	101	89	102	125	111	110
	PrecMax	1685	172	218	382	245	366	327	191	292	260	391	195	182
	PrecMin	883	13	28	24	7	21	28	21	6	37	5	49	35
	PrecStd	207	35	42	73	62	75	67	50	59	63	86	38	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	14.1	21.5	28.4	31.4	32.9	33.7	33.4	32.1	25.7	19.9	14.2
		MCWB	11.7	9.0	12.8	16.1	19.8	23.1	24.0	23.8	22.9	18.2	15.0	11.1
	2%	DB	8.9	9.9	15.2	23.1	27.7	30.1	31.8	31.1	28.7	22.6	16.6	10.7
		MCWB	6.1	6.1	9.5	14.5	17.9	21.6	22.8	22.4	21.0	16.9	12.9	8.4
	5%	DB	6.1	7.1	11.4	19.4	24.8	27.8	30.1	29.1	26.3	19.7	14.0	8.2
		MCWB	3.7	3.7	6.4	12.3	16.3	20.0	21.9	21.6	19.6	15.7	10.9	5.9
	10%	DB	3.6	4.1	8.6	16.5	22.0	26.0	28.4	27.5	23.9	17.3	12.0	6.0
		MCWB	1.3	1.4	4.7	9.7	15.3	18.6	21.3	20.5	18.6	14.0	9.2	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.2	10.8	14.3	18.0	21.4	24.5	25.4	25.3	23.9	20.5	16.3	12.5
		MCDB	12.8	12.4	20.5	23.6	29.4	30.3	31.3	31.1	28.4	22.9	18.1	13.7
	2%	WB	7.0	6.7	10.6	15.6	19.4	22.9	24.3	23.8	22.3	18.0	13.9	8.9
		MCDB	8.1	9.1	14.8	22.1	25.1	28.6	29.6	28.6	27.0	20.6	15.5	10.1
	5%	WB	3.9	3.9	7.4	13.2	17.8	21.6	23.3	22.8	21.1	16.4	11.5	6.4
		MCDB	5.8	6.3	10.0	18.6	22.7	26.0	28.2	26.9	24.5	19.0	13.4	7.8
	10%	WB	1.6	2.0	5.1	10.4	16.0	20.1	22.3	21.8	19.8	14.5	9.5	4.0
		MCDB	3.2	4.1	8.3	15.3	20.8	23.9	26.7	25.6	22.9	16.9	11.7	5.7
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.4	9.7	11.2	11.3	11.0	11.2	11.6	11.5	10.1	9.0	7.9
		MCDBR	10.3	12.1	13.7	16.6	16.3	14.0	13.1	13.5	13.6	13.2	12.2	10.5
		MCWBR	9.0	9.1	9.0	9.3	8.0	6.9	6.0	6.2	7.2	8.3	9.2	8.9
	5% WB	MCDBR	10.0	11.6	12.7	15.1	14.2	12.5	11.3	11.4	11.5	11.4	10.2	10.1
		MCWBR	9.2	9.5	9.1	9.7	7.9	7.0	5.8	5.7	7.0	8.0	8.9	9.4
Clear-Sky Solar Irradiance	taub	0.282	0.294	0.318	0.351	0.388	0.434	0.453	0.421	0.369	0.331	0.310	0.289	
	taud	2.462	2.439	2.464	2.426	2.362	2.259	2.213	2.291	2.417	2.529	2.546	2.529	
	Ebn at Noon	862	910	927	915	885	841	819	835	858	857	824	824	
	Edh at Noon	72	87	97	110	120	134	139	124	101	80	67	63	
All-Sky Solar Radiation	RadAvg	1.70	2.57	3.65	4.60	5.12	5.51	5.79	5.26	4.24	2.73	1.82	1.36	
	RadStd	0.11	0.20	0.23	0.50	0.59	0.48	0.38	0.26	0.25	0.25	0.17	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	-0.75	N/A	N/A	N/A	N/A
Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+82	+31

Nomenclature: See separate page